Simple Object Detection and Tracking Labs

By YOUSSOUFA Mohamadou

Get Code from Github Repo

A tutorial for object detection, face detection, and face recognition. You will run three Jupyter notebooks in **VS Code** on a **CPU** (no GPU required).

Lab	Notebook	What you will do
Lab 01	`Lab01_Face Detection in Images.ipynb`	Detect objects and faces in still images with YOLO
Lab 02	`Lab02_Face Detection in Live Videos.ipynb`	Run the same detection on a live camera feed
Lab 03	`Lab03_Face_Recognition.ipynb`	Train an OpenCV LBPH recognizer on your own photos and recognize faces in real time

Labs 02 and 03 need a webcam (built-in or USB).

What you need

1. Python 3.12
2. Visual Studio Code
3. These VS Code extensions (install from the Extensions view):

- Python
- Jupyter

CPU setup with uv

Open a terminal in the project folder (in VS Code: **Terminal** → **New Terminal**).

Create and activate a virtual environment:

```
python -m venv .venv
```

- Linux / macOS: `source .venv/bin/activate`
- Windows: `.venv\Scripts\activate`

Install **uv**, then the CPU PyTorch wheels, YOLO, and OpenCV. Ultralytics installs `opencv-python`; uninstall it and replace it with `opencv-contrib-python` so Lab 03 can use the LBPH face recognizer.

```
pip install uv
uv pip install torch torchvision --index-url
```

```
https://download.pytorch.org/whl/cpu
uv pip install ultralytics
uv pip uninstall opencv-python
uv pip install opencv-contrib-python
```

Install the remaining packages:

Install the packages you need if you do not have them already :

```
uv pip install numpy
uv pip install Pillow
uv pip install ipython
uv pip install ipykernel
uv pip install jupyter
```

Run the labs in VS Code

1. Open this folder in VS Code (**File → Open Folder**).
2. Open one of the three `.ipynb` lab files.
3. Click **Select Kernel** in the top right of the notebook.
4. Choose the `.venv` interpreter you just created.
5. Run the cells from top to bottom.

The first time you run a YOLO cell, the model file is downloaded automatically into `yolo_models/`.

Lab presentations

Each lab notebook can be presented as slides in VS Code.

1. Open the lab notebook.
2. Select the `.venv` kernel.
3. Open the Command Palette (`Ctrl+Shift+P` / `Cmd+Shift+P`) and run **View as Slide Show**.

Lab	Notebook	What the slides cover
Lab 01	`Lab01_Face Detection in Images.ipynb`	Object detection with the official YOLO model, then face detection on still images
Lab 02	`Lab02_Face Detection in Live Videos.ipynb`	The same detection pipeline on a live camera feed
Lab 03	`Lab03_Face_Recognition.ipynb`	Training an OpenCV LBPH recognizer on your own photos and recognizing faces in real time

To export a standalone HTML slideshow (optional):

```
jupyter nbconvert "Lab01_Face Detection in Images.ipynb" --to slides
jupyter nbconvert "Lab02_Face Detection in Live Videos.ipynb" --to slides
```

```
jupyter nbconvert "Lab03_Face_Recognition.ipynb" --to slides
```

Each command writes a `.slides.html` file next to the notebook. Open that file in a browser.

Face detection model source

This workflow uses official YOLO models downloaded from Python code, plus a community face model, `yolov8m-face.pt`, from akanametov/yolo-face.

Please give Akanametov a star if you use this workflow: https://github.com/akanametov

Demo images

Clone this repository. The resized demo images are already included and ready to use.

If you prefer the original high-resolution files, download them from Magnific and resize them yourself.

- demo.jpg (Girl + Cat): Download
- demo1.jpg (Cyclist + Cars): Download
- demo2.jpg (Kitchen): Download
- demo3.jpg (Dog + Sheep): Download
- demo4.jpg (Laptop, Cup & Mouse): Download

Resize each downloaded image to about 15% of its original size:

```
import cv2

img = cv2.imread("original_image_name.jpg")

resized_img = cv2.resize(
    img,
    None,
    fx=0.15,
    fy=0.15
)

cv2.imwrite("resized_image_name.jpg", resized_img)
```

Acknowledgements

These tutorials were made thanks to resources shared by:

1. MariyaSha here
2. Akanametov here